

Smart Accident Alert System Using GSM Sensor

An innovative system to speed up accident response using GSM sensors.
This technology aims to save lives through timely alerts.

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Introduction: The Problem of Delayed Accident Response

Delay Causes

Location detection difficulties and communication lags hinder response.

Impact

Every minute lost reduces survival chances significantly.

Need

Fast, automatic alerts can improve emergency response times.

System Architecture: GSM Sensor and Microcontroller Integration

GSM Sensor

Detects accident signals and enables mobile communication.

Microcontroller

Processes sensor data and controls alert transmission.

Integration

Seamless communication between sensors and GSM ensures quick alerts.

Sensor Data Acquisition and Processing

Data Capture

Accelerometer detects sudden impacts.

Signal Filtering

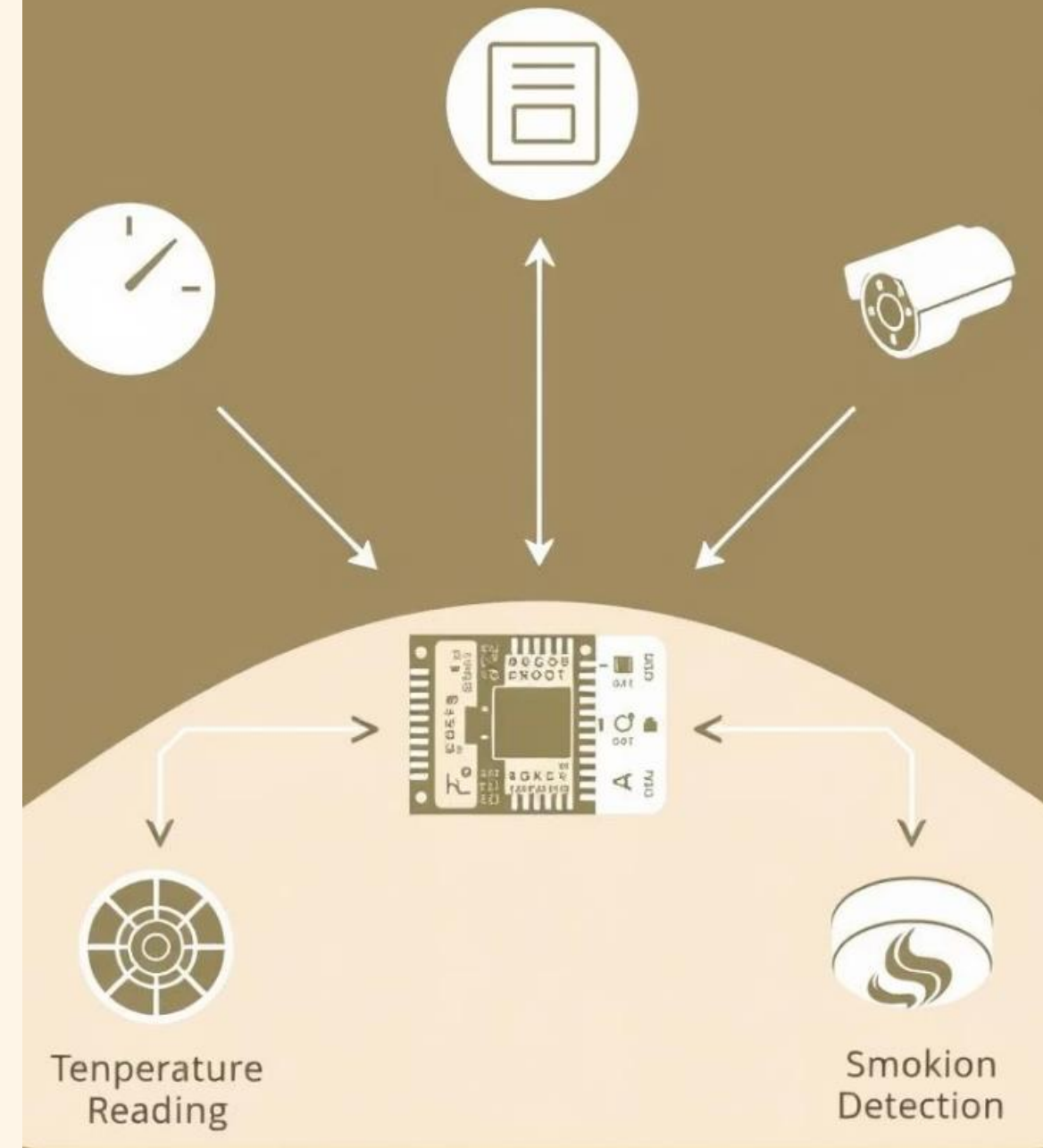
Noises are minimized for accurate detection.

Threshold Check

Evaluates if data confirms an accident event.

Trigger Alert

Initiates emergency communication via GSM.



Advantages

1. Rapid Emergency Response: Sends alerts instantly via GSM, reducing emergency response time by over 80%.
2. Life-Saving Potential: Increases survival chances by minimizing delays after an accident.
3. Automation: Fully automated; no manual action needed during emergencies.
4. Wide Network Compatibility: GSM module can communicate over existing mobile networks.
5. Scalability: Can be integrated with GPS, AI, and smart city infrastructure for enhanced capabilities.

GSM Module Communication Protocol

Initialization

Module registers with the cellular network automatically.

Message Composition

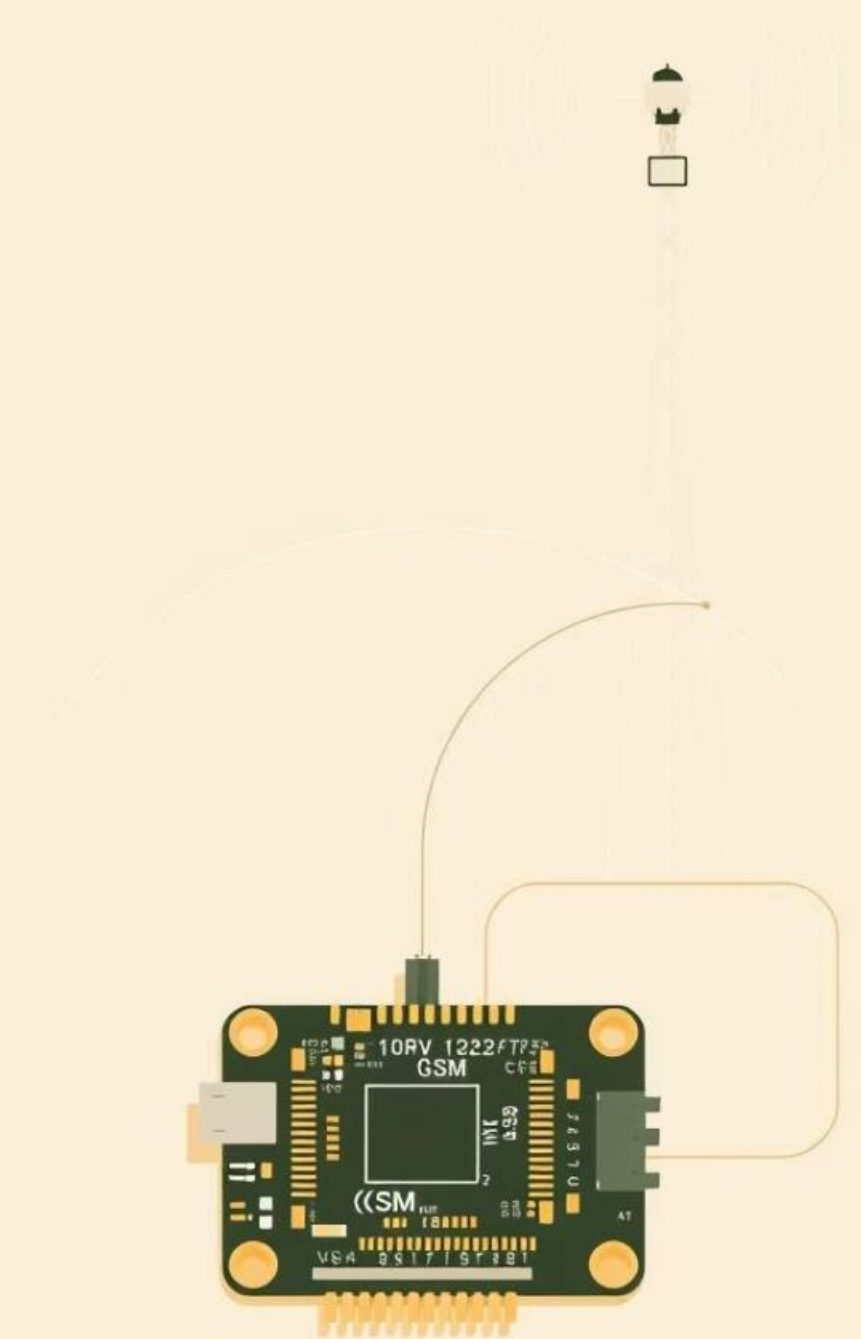
Accident details like location and time formatted as SMS.

Transmission

SMS sent to emergency contacts or control center.

Acknowledgment

System waits for delivery confirmation to ensure alert reception.



Prototype Design and Implementation



Microcontroller

Handles sensor data and logic processing.



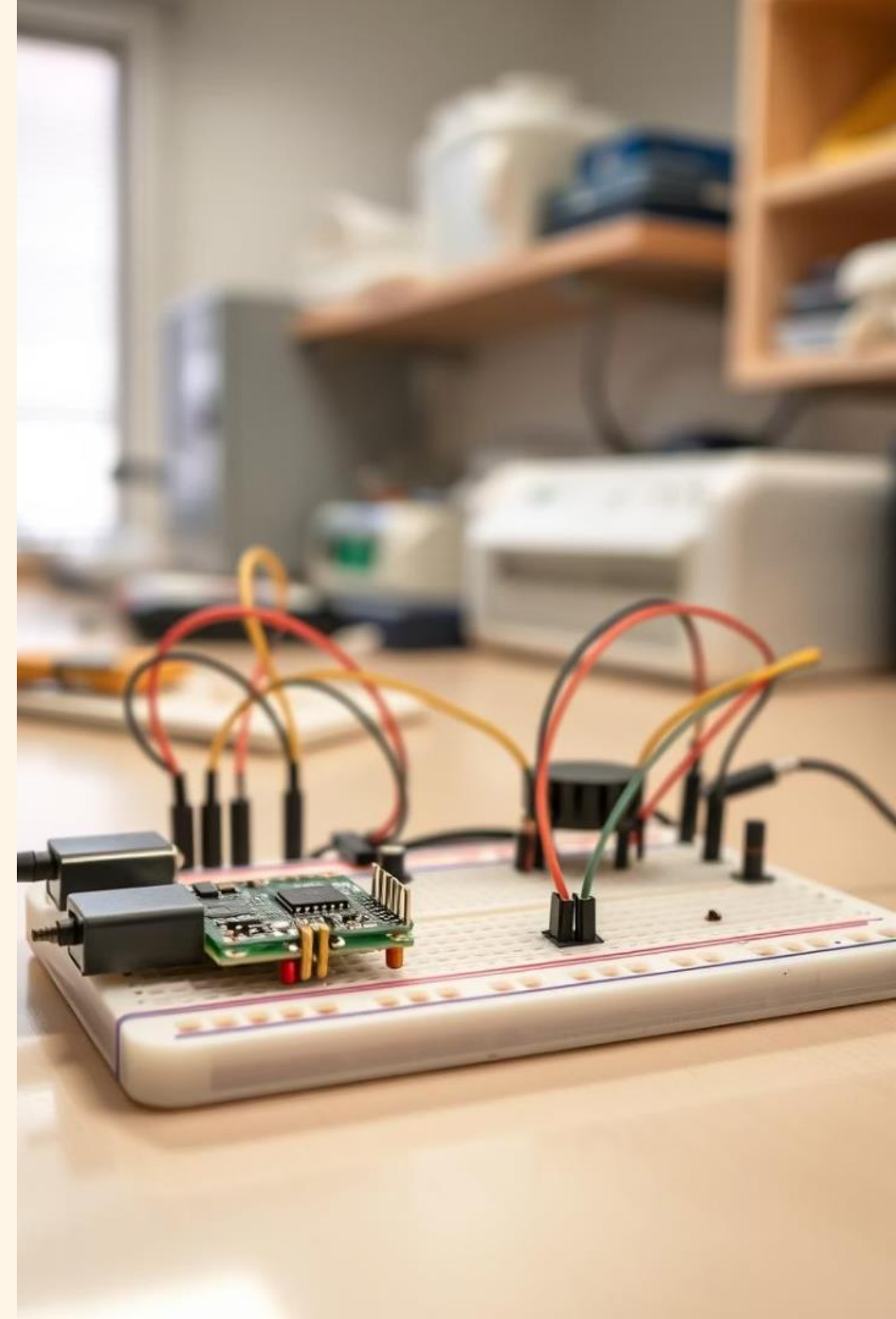
GSM Module

Enables wireless communication across mobile networks.

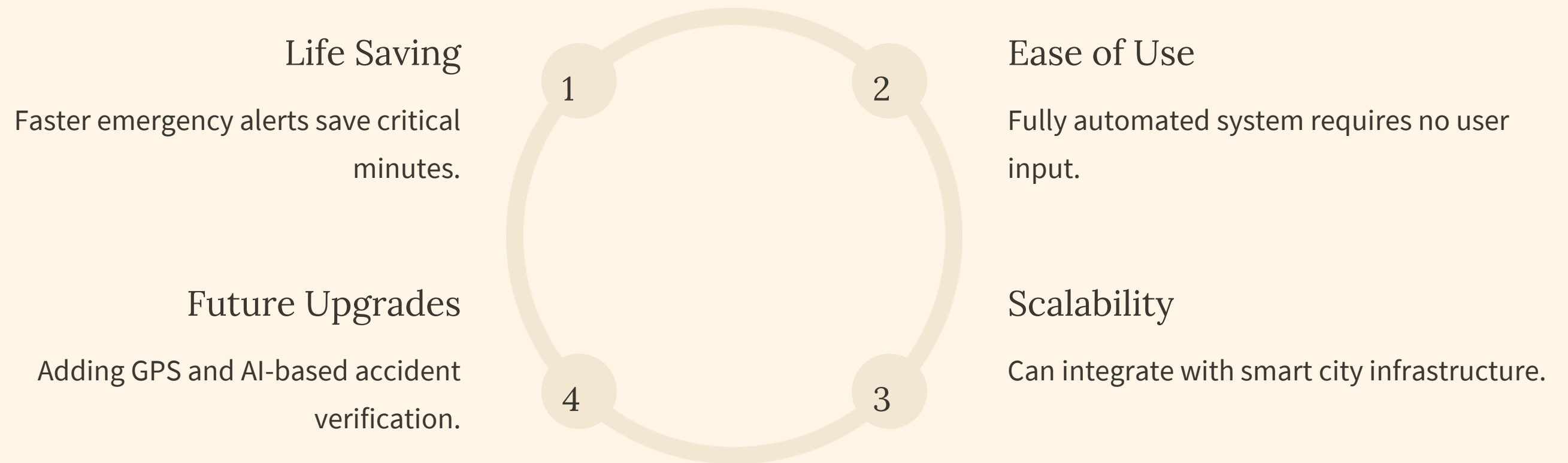


Power Supply

Ensures reliable operation under varying conditions.



Conclusion: Benefits and Future Enhancements



THANK YOU